

1 Mapping the QM-IDI in HYDRA

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Features of the QM-IDI

In the context of a connection via the QM-IDI interface, it is the task of HYDRA to receive inspection processing data and to integrate them into the HYDRA process. In addition to that, HYDRA creates the appropriate data from recorded confirmations and transfers these to SAP QM. Both scenarios are initiated by HYDRA.

SAP QM → HYDRA – movement and master data

Download of inspection data

The transfer of inspection batch data from SAP to HYDRA will be carried out using the function module QIRF_SEND_REQUIREMENTS_GET_DAT2. For that purpose, HYDRA transfers selection options to the function module. Those options are transferred using the structure QAILS. Additionally, the function module offers the possibility to control the download in detail. The content of each parameter can be configured in HYDRA in customizing table [SAP FM PARAM CFG](#).

Parameter name:	T	L	D	Description
I_IND_SORT_ASCENDING_TO_DATE	CHAR	1		Indicator: Sort operations in ascending order according to creation date Default = "X"
I_IND_SORT_DESCENDING_TO_DATE	CHAR	1		Indicator: Sort operations in descending order according to creation date Default= " "
I_IND_MULTI_TRANSFER_POSSIBLE	CHAR	1		Indicator: Correction transfers of operations possible, Default = " "
I_IND_EVALUATE_CHARACTERISTIC	CHAR	1		Indicator: Transfer valuation specifications for characteristics, Default = "X"
I_IND_SET_BLOCK_INDICATORS	CHAR	1		Indicator: Set lock entries in QM, Default = "X"
I_IND_ONLY_OBLIGATORY_CHARACT	CHAR	1		Indicator: Only transfer required characteristics, Default = " "
I_IND_TRANSFER_CHAR_CODES	CHAR	1		Indicator: Transfer catalog data for inspection characteristics, Default = "X"

Parameter name:	T	L	D	Description
I_IND_READ_WORK_CENTER	CHAR	1		Indicator: Read work center data, Default = "X"
I_IND_READ_VENDOR_AND_PRODUCER	CHAR	1		Indicator: Read vendor data, Default = "X"
I_IND_READ_PURCHASING_INFO				Indicator: Read data from purchasing information record, Default = "X"
I_IND_READ_SALES_INFO				Indicator: Read data from sales information record, Default = "X"
I_IND_SEND_PROTOCOL_MAIL				Indicator: Send error log by mail Default = "X"
I_IND_TRANSFER_USAGE_DEC_CODES				Indicator: Transfer catalog data for usage decision, Default = "X"

The structure QAILS contains the selection options. The content depends on the business scenario and will be described separately. It has the following structure:

Field	T	L	Description
SATZART	CHAR	3	Record type for request record
LOSNR_VON	NUMC	12	From inspection batch number
LOSNR_BIS	NUMC	12	To inspection batch number
PLNFL	CHAR	6	Operation sequence in task list
VORNR_VON	CHAR	4	From operation number
VORNR_BIS	CHAR	4	To operation number
VORGWERK	CHAR	4	Plant of operation to be processed
SUBSYS	CHAR	6	Identifier of the subsystem
PRPLATZ	CHAR	8	Work center
PRPLATZWRK	CHAR	4	Plant of the work center
MATNR	CHAR	18	Material number
DATUM_VON	DATE	8	From creation date of inspection batch
DATUM_BIS	DATE	8	To creation date of inspection batch
PRUEFSTAT	CHAR	1	Status of the inspection
ART	CHAR	8	Inspection type
HERKUNFT	CHAR	2	Origin of the inspection batch
CHARG	CHAR	10	Batch number
AUFNR_VON	CHAR	12	From order number
AUFNR_BIS	CHAR	12	To order number
LIFNR	CHAR	10	Vendor number

Field	T	L	Description
KUNNR	CHAR	10	Customer number

According to the selection option, the function module provides the inspection batch data, including inspection batch header, operation and characteristics. The structures will be explained in detail in chapter **Error! Reference source not found. Error! Reference source not found.**, also pointing out the meaning of the different fields.

Besides the inspection data, the function module also provides a table of error messages. These error messages will be displayed in the MLE inbound transactions and in the protocol of the communication program. Additionally, those entries are also forwarded to the HYDRA Escalation Management (Prerequisite: license SIS-ESK and SAP-ESK) with escalation SAP.QM_IDI_INBOUND_MSG.

The download of inspection batch data will be carried out by the hysapqmc.exe/out program.

Download catalog master data

The transfer of catalog master data from SAP to HYDRA will be carried out by using the function module QIRF_SEND_CATALOG_DATA2. For that purpose, HYDRA transfers selection options to the function module. The content of each parameter can be configured in HYDRA in customizing table [SAP_FM_PARAM_CFG](#).

Parameter name:	T	L	D	Description
I_IND_CATALOG_IS_SEL_SET	CHAR	1		Indicator: Choose selected set, Default = " "
I_IND_CATALOG_IS_CODEGROUP	CHAR	1		Indicator: Choose code groups Default = " "
I_IND_CATALOG_TYPE	CHAR	3		Catalog type Default = "3"
I_IND_PLANT_OF_SELECTED_SET	CHAR	4		Plant of selected set Default = "0001"
I_IND_CATALOG	CHAR	4		Catalog (selected set or code group)
I_IND_LANGUAGE	CHAR	2		Language Default = "D"

Besides the catalog data, the function module also provides a table of error messages. These error messages will be displayed in the MLE inbound transactions, the protocol of the communication program. Additionally, those entries are also forwarded to the HYDRA Escalation Management (Prerequisite: license SIS-ESK and SAP-ESK) with escalation SAP.QM_IDI_INBOUND_MSG.

The download of catalog master data will be carried out by the hysapqmc.exe/out program.

Download inspection point data

The transfer of inspection point data from SAP to HYDRA will be carried out by using the function module QIRF_INSPPOINT_GETLIST. For that purpose, HYDRA transfers selection options to the function module. The content of each parameter can be configured in HYDRA in customizing table [SAP FM PARAM_CFG](#).

Basically there are two options for downloading inspection points from SAP

- HYDRA-triggered e.g. for stand-alone-CAQ

In this case the selection option are configured statically in HYDRA.

- Dynamically-triggered based on the inspection data download

In this case the inspection lot number is determined from downloaded data and used to select the inspection points inspections lot related.

Parameter name:	T	L	D	Description
INSPLOT	-	-		No usage
INSPOPER	CHAR	4		
INSPPOINT_FROM	NUMC	6		Inspection point from Default = "000001"
INSPPOINT_TO	NUMC	6		Inspection point to Default = "999999"
I_SUBSYS	CHAR	6		Subsystem confirms results Default = „QM0001“

Besides the inspection point data, the function module also provides a table of error messages. These error messages will be displayed in the MLE inbound transactions, the protocol of the communication program. Additionally, those entries are also forwarded to the HYDRA Escalation Management (Prerequisite: license SIS-ESK and SAP-ESK) with escalation SAP.QM_IDI_INBOUND_MSG.

The download of inspection point master data will be carried out by the hysapqmc.exe/out program.

HYDRA → SAP QM – inspection results / inspection points

HYDRA records results according to the inspection specifications. Those are to be transferred to SAP QM. For that purpose, SAP provides a set of function modules. According to the record type of the recorded inspection results one or the other has to be used.

Upload for Single Results

The QIRF_GET_ORIGINAL_VALUES2 function module has to be used to upload single results for inspections. Besides the QAISE structure, which carries the actual inspection results, the function module offers additional parameters that allow controlling the posting process in SAP QM. The content of each parameter can be configured in HYDRA in customizing table [SAP FM PARAM CFG](#).

Parameter name:	T	L	D	Description
I_IND_EVALUATION_TRANSFER	CHAR	1		Indicator: Transfer valuations from subsystem Default = "X"
I_IND_CLOSE_PROCESSING	CHAR	1		Indicator: Close samples or characteristics, Default = " "
I_SEND_PROTOCOL_MAIL	CHAR	1		Indicator: Send error log by mail, Default = "X"
I_SUBSYS	CHAR	6		Subsystem confirms results Default = „QM0001“
I_IND_POSTING_KZ	CHAR	1		Indicator: Call up program for updating results, Default = "X"
I_IND_PROC_COMMIT_WORK	CHAR	1		Indicator.: Trigger commit work to database, Default = "X"

As a result of the posting process in SAP QM, the function module provides a table of error messages. These messages will be displayed in the MLE outbound transactions in the protocol of the communication program. Additionally, those entries are also forwarded to the HYDRA Escalation Management (Prerequisite: license SIS-ESK and SAP-ESK) with escalation SAP.QM_IDI_OUTBOUND_MSG.

Upload for Sample Results

The QIRF_GET_SAMPLE_VALUES2 function module has to be used to upload sample results for inspections. Besides the QAISR structure, which carries the actual inspection results, the function module offers additional parameters that allow controlling the posting process in SAP QM. The content of each parameter can be configured in HYDRA in customizing table [SAP FM PARAM CFG](#).

Parameter name:	T	L	D	Description
I_IND_EVALUATION_TRANSFER	CHAR	1		Indicator: Transfer valuations from subsystem Default = "X"
I_IND_CLOSE_PROCESSING	CHAR	1		Indicator: Close samples or characteristics, Default = " "

Parameter name:	T	L	D	Description
I_SEND_PROTOCOL_MAIL	CHAR	1		Indicator: Send error log by mail, Default = "X"
I_SUBSYS	CHAR	6		Subsystem confirms results Default = „QM0001“
I_IND_POSTING_KZ	CHAR	1		Indicator: Call up program for updating results, Default = "X"
I_IND_PROC_COMMIT_WORK	CHAR	1		Indicator.: Trigger commit work to database, Default = "X"

As a result of the posting process in SAP QM, the function module provides a table of error messages. These error messages will be displayed in the MLE outbound transactions in the protocol of the communication program. Additionally, those entries are also forwarded to the HYDRA Escalation Management (Prerequisite: license SIS-ESK and SAP-ESK) with escalation SAP.QM_IDI_OUTBOUND_MSG.

Upload for Characteristic Results

The QIRF_GET_FEATURE_VALUES2 function module has to be used to upload characteristic results for inspections. Besides the QAIMR structure, which carries the actual inspection results, the function module offers additional parameters that allow controlling the posting process in SAP QM. The content of each parameter can be configured in HYDRA in customizing table [SAP FM PARAM CFG](#).

Parameter name:	T	L	D	Description
I_IND_EVALUATION_TRANSFER	CHAR	1		Indicator: Transfer valuations from subsystem Default = "X"
I_IND_CLOSE_PROCESSING	CHAR	1		Indicator: Close samples or characteristics, Default = " "
I_SEND_PROTOCOL_MAIL	CHAR	1		Indicator: Send error log by mail, Default = "X"
I_SUBSYS	CHAR	6		Subsystem confirms results Default = „QM0001“
I_IND_POSTING_KZ	CHAR	1		Indicator: Call up program for updating results, Default = "X"
I_IND_PROC_COMMIT_WORK	CHAR	1		Indicator.: Trigger commit work to database, Default = "X"

As a result of the posting process in SAP QM, the function module provides a table of error messages. These error messages will be displayed in the MLE outbound transactions in the protocol of the communication program. Additionally, those entries are also forwarded to the HYDRA Escalation Management (Prerequisite: license SIS-ESK and SAP-ESK) with escalation SAP.QM_IDI_OUTBOUND_MSG.

Upload for Inspection Points

Creating and updating / deleting inspection points are done by using the QIRF_GET_INSP_POINT2 function module. Besides the QAIPP structure, which carries the actual inspection point data, the function module offers additional parameters that allow controlling the posting process in SAP QM. The content of each parameter can be configured in HYDRA in customizing table [SAP FM PARAM CFG](#).

Parameter name:	T	L	D	Description
I_SEND_PROTOCOL_MAIL	CHAR	1		Indicator: Send error log by mail, Default = "X"
I_SUBSYS	CHAR	6		Subsystem confirms results Default = „QM0001“

As a result of the posting process in SAP QM, the function module provides a table of error messages. These error messages will be displayed in the MLE outbound transactions in the protocol of the communication program. Additionally, those entries are also forwarded to the HYDRA Escalation Management (Prerequisite: license SIS-ESK and SAP-ESK) with escalation SAP.QM_IDI_OUTBOUND_MSG.

Upload for Usage Decision

The usage decision is uploaded by using the QIRF_GET_USAGE_DECISION2 function module. Besides the QAIVE structure, which carries the actual usage decision data, the function module offers additional parameters that allow controlling the posting process in SAP QM. The content of each parameter can be configured in HYDRA in customizing table [SAP FM PARAM CFG](#).

Parameter name:	T	L	D	Description
I_STOCK_POSTING	CHAR	1		Indicator: Trigger inventory posting in QM after usage decision Default = "X"
I_SEND_PROTOCOL_MAIL	CHAR	1		Indicator: Send error log by mail, Default = "X"
I_SUBSYS	CHAR	6		Subsystem confirms results Default = „QM0001“

As a result of the posting process in SAP QM, the function module provides a table of error messages. These error messages will be displayed in the MLE outbound transactions in the protocol of the communication program. Additionally, those entries are also forwarded to the HYDRA Escalation Management (Prerequisite: license SIS-ESK and SAP-ESK) with escalation SAP.QM_IDI_OUTBOUND_MSG.

Upload for Defect Items

The defect items are uploaded by using the QIRF_GET_DEFECT_ITEMS2 function module. Besides the QMIFE structure, which carries the actual defect items data, the function module offers additional parameters that allow controlling the posting process in SAP QM. The content of each parameter can be configured in HYDRA in customizing table [SAP_FM_PARAM_CFG](#).

Parameter name:	T	L	D	Description
I_SEND_PROTOCOL_MAIL	CHAR	1		Indicator: Send error log by mail, Default = "X"
I_SUBSYS	CHAR	6		Subsystem confirms results Default = „QM0001“

As a result of the posting process in SAP QM, the function module provides a table of error messages. These error messages will be displayed in the MLE outbound transactions in the protocol of the communication program. Additionally, those entries are also forwarded to the HYDRA Escalation Management (Prerequisite: license SIS-ESK and SAP-ESK) with escalation SAP.QM_IDI_OUTBOUND_MSG.

Supported integration scenarios

Each function module provided on the SAP side expects specific import parameters, structures or tables. Depending on the function module, the data in these ones are derived either from configuration in HYDRA or from integration with other interfaces. The implementation of the QM-IDI in HYDRA serves the following processes and scenarios:

- Integration with SAP standard interface PP-PDC
- Integration with SAP standard interface PP-PDC and HYDRA interface extension HYINFO
- Stand-alone CAQ (e.g. for goods receipt)

Integration with SAP standard interface PP-PDC

The integration between QM-IDI and PP-PDC interface serves the requirements of increased in process control during production process. For that reason the routing in SAP not only contains productive operation (transferred via PP-PDC interface) but also QM operations that itself contain characteristics and which are transferred using QM-IDI interface. It is also possible that an operation is transferred via both interfaces.

The download process is started by releasing the production order in SAP. That causes a download of the production order and its operations via PP-PDC interface. The data will be received in HYDRA in the MES Link Enabling layer for further processing. Instead of performing the posting into the HYDRA application tables, the data is used to receive the inspection data. For that specific data of the PP-PDC IDoc E2BP-PP-PDC-OPERA2000 is taken and transferred into the QM-IDI QAILS structure to request the inspection batch for each order. The following table indicates which fields of the PP-PDC IDoc are used to request inspection specifications:

Field	Type	L	Description	Usage / Origins
SATZART	CHAR	3	Record type to request record	Fix „Q40“
LOSNR_VON	NUMC	12	From inspection batch number	Not used
LOSNR_BIS	NUMC	12	To inspection batch number	Not used
PLNFL	CHAR	6	Operation sequence in task list	Not used
VORNR_VON	CHAR	4	From operation number	Not used
VORNR_BIS	CHAR	4	To operation number	Not used
VORGWERK	CHAR	4	Plant of operation to be processed	E2BP_PP_PDC_OPERA2000. PLANT
SUBSYS	CHAR	6	Identifier of the subsystem	Customizing table in HYDRA
PRPLATZ	CHAR	8	Work center	Not used
PRPLATZWRK	CHAR	4	Plant of the work center	Not used
MATNR	CHAR	18	Material number	E2BP_PP_PDC_OPERA2000. MATERIAL
DATUM_VON	DATE	8	From creation date of inspection batch	Not used
DATUM_BIS	DATE	8	To creation date of inspection batch	Not used
PRUEFSTAT	CHAR	1	Status of the inspection	Not used
ART	CHAR	8	Inspection type	Not used
HERKUNFT	CHAR	2	Origin of the inspection batch	Not used
CHARG	CHAR	10	Batch number	Not used
AUFNR_VON	CHAR	12	From order number	E2BP_PP_PDC_OPERA2000. ORDERID
AUFNR_BIS	CHAR	12	To order number	E2BP_PP_PDC_OPERA2000. ORDERID
LIFNR	CHAR	10	Vendor number	Not used

KUNNR	CHAR	10	Customer number	Not used
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As a result of the download of the inspection specification the original PP-PDC IDoc and the newly transferred QM-IDI data are pooled into a new IDoc of the following characteristics:

Message type / IDoc type: PPCC2RECORDER_QM_IDI

Segments:	E2PPCC2RECORDER (Indicator initial download)	1
	E2BP_PP_PDC_OPERA2000 (PP operations)	1-n
	E2BP_PP_PDC_OPERA1000 (Deletion of PP operations.)	0-n
	Z2QIAVC000X000 (Inspection batches and operations)	1-n
	Z2QAIMV000X000 (inspection characteristics)	1-n
	Z2QAICA000X000 (catalogs)	1-n
	Z2QIERR000X000 (error logs)	1-n

Prerequisite:

- HYDRA license SAP-PPDC

Integration with SAP PP-PDC and HYINFO

The integration of QM-IDI with PP-PDC can also be extended when using HYDRA interface extension for PP-PDC HYINFO. The interface extension HYINFO also uses sRFC calls to receive additional data from SAP such as order header data or the list of components.

In this case, the original PP-PDC will be used first to collect additional data within SAP from the interface extension. As a result the interface extension pools the original order data from PP-PDC interface and the newly selected additional data into a new IDoc of the following characteristics:

Message type / IDoc type: PPCC2HYINFOORDER

Segments:	E2PPCC2RECORDER (Indicator initial download)	1
	E2BP_PP_PDC_OPERA2000 (PP operations)	1-n
	E2BP_PP_PDC_OPERA1000 (Deletion of PP operations.)	0-n
	HYINFO_xxx (additional data from interface extension)	1-n

This new IDoc is the basement to collect inspection specifications in SAP QM. For that, the original PP-PDC data will be used to derive the production orders to request the specifications for. For that the same data / fields will be used as described above. As a result a new IDoc will be created by pooling as much the original PP-PDC data, the data from interface extension HYINFO as the QM-IDI data with the following characteristics:

Message type / IDoc type: PPCC2HYINFOORDER_QM_IDI

Segments:	E2PPCC2RECORDER (Indicator initial download)	1
	E2BP_PP_PDC_OPERA2000 (PP operations)	1-n
	E2BP_PP_PDC_OPERA1000 (Deletion of PP operations.)	0-n
	HYINFO_xxx (additional data from interface extension)	1-n
	Z2QIAVC000X000 (Inspection batches and operations)	1-n
	Z2QAIMV000X000 (inspection characteristics)	1-n
	Z2QAICA000X000 (catalogs)	1-n
	Z2QIERR000X000 (error logs)	1-n

Stand-alone CAQ

The stand-alone variant is thought to support inspections that are not related to production orders, such as goods receipt inspections. It can be used independently or in combination with the two integration scenarios described in the previous chapters.

In order to be able to request the inspection batches for those inspections, there is a customizing table in HYDRA that allows defining different selection options. To support different kinds of selection options it is possible to configure multiple variants in the customizing table [SAP_FB_PARAM_CFG](#).

The request of inspection batch will be performed on a cyclic base. For the calling program hysapqmc.exe/out has to be configured in the HYDRA Scheduler.

As a result of the receipt of inspection batches an IDoc will be created with the following characteristics:

Message type / IDoc type: ZQM_IDI

Segments:	Z2QIAVC000X000 (Inspection batches and operations)	1-n
	Z2QAIMV000X000 (inspection characteristics)	1-n
	Z2QAICA000X000 (catalogs)	1-n
	Z2QIERR000X000 (error logs)	1-n